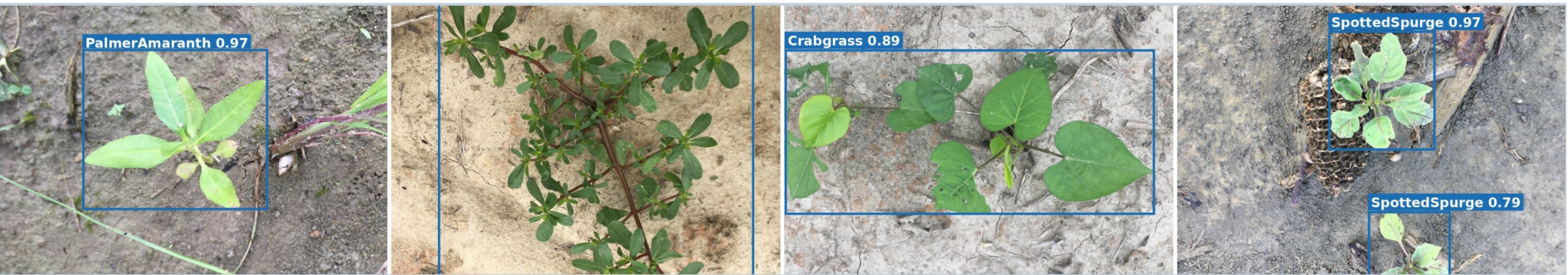


Built at the MTSU Great Robotics Lab. Every robot the lab owns reports into one platform; agents on it harvest, filter, label and train; a third agent watches the other two. This is what runs today, what the robots have recorded, and what six months of running it taught us.

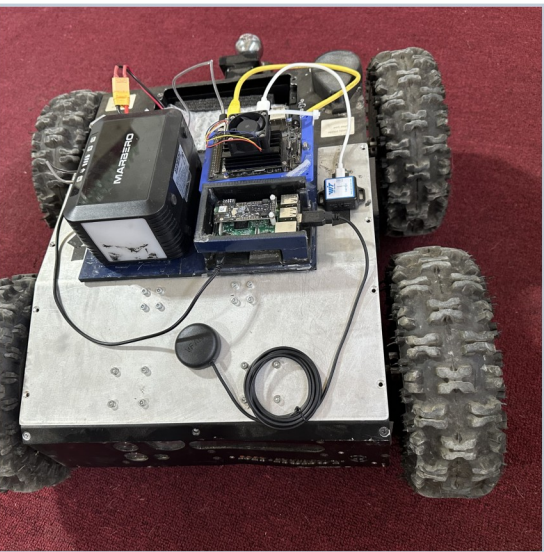
WHAT EACH PROJECT IS, AND WHAT IT IS FOR

Weed Detection



Two agents work the same corpus without being asked. One goes out to public sources, brings weed imagery back and audits what it brought; the other trains a detector on whatever survives the audit and scores it. The registry behind them holds more than 150,000 labelled weed images, and the detector that comes out is small enough to ride on the robots themselves.

241 Robot



A tracked rover built for the row: narrow enough to turn inside one and tough enough to cross the beds between them, so it can reach parts of a field a tractor cannot. It carries a camera, GPS and two inertial units and streams all of them live into the platform. That stream is what an autonomous-driving model for the field is trained on, and a signed-in user drives the rover from a plain browser.

Laser Cart



A twin-laser weeder. A YOLO detector finds the weed in the frame, the two galvanometer lasers are aimed at it and fired, and the weed is killed where it stands -- no chemical, and nothing touches the crop beside it. One camera looks down over the work zone and one looks forward, and both stream into the platform while the cart's own operator drives and fires it.

Humanoid and Quadruped



A humanoid and a quadruped, taken out to the farm to record where wheels do not go. Their project is already open and takes the same ingest contract the rover and the cart use, so the day either of them opens a session it is a governed dataset like any other.

Figure 1. Every project on the platform was opened the same way, sends through the same ingest contract, and is trained and evaluated by the same agents.

HOW A ROUND RUNS, AND WHAT IT DOES WHEN IT IS STUCK

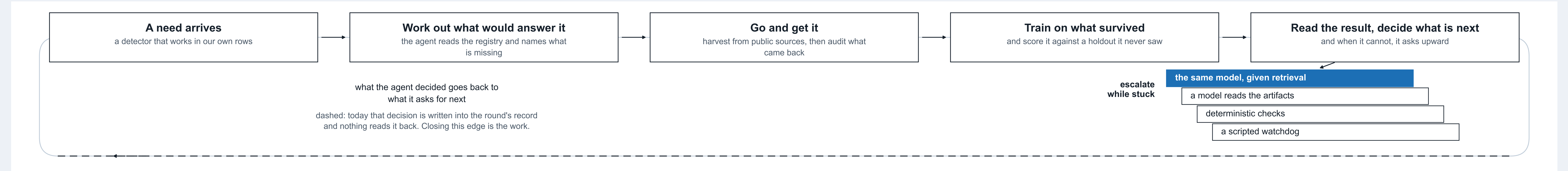
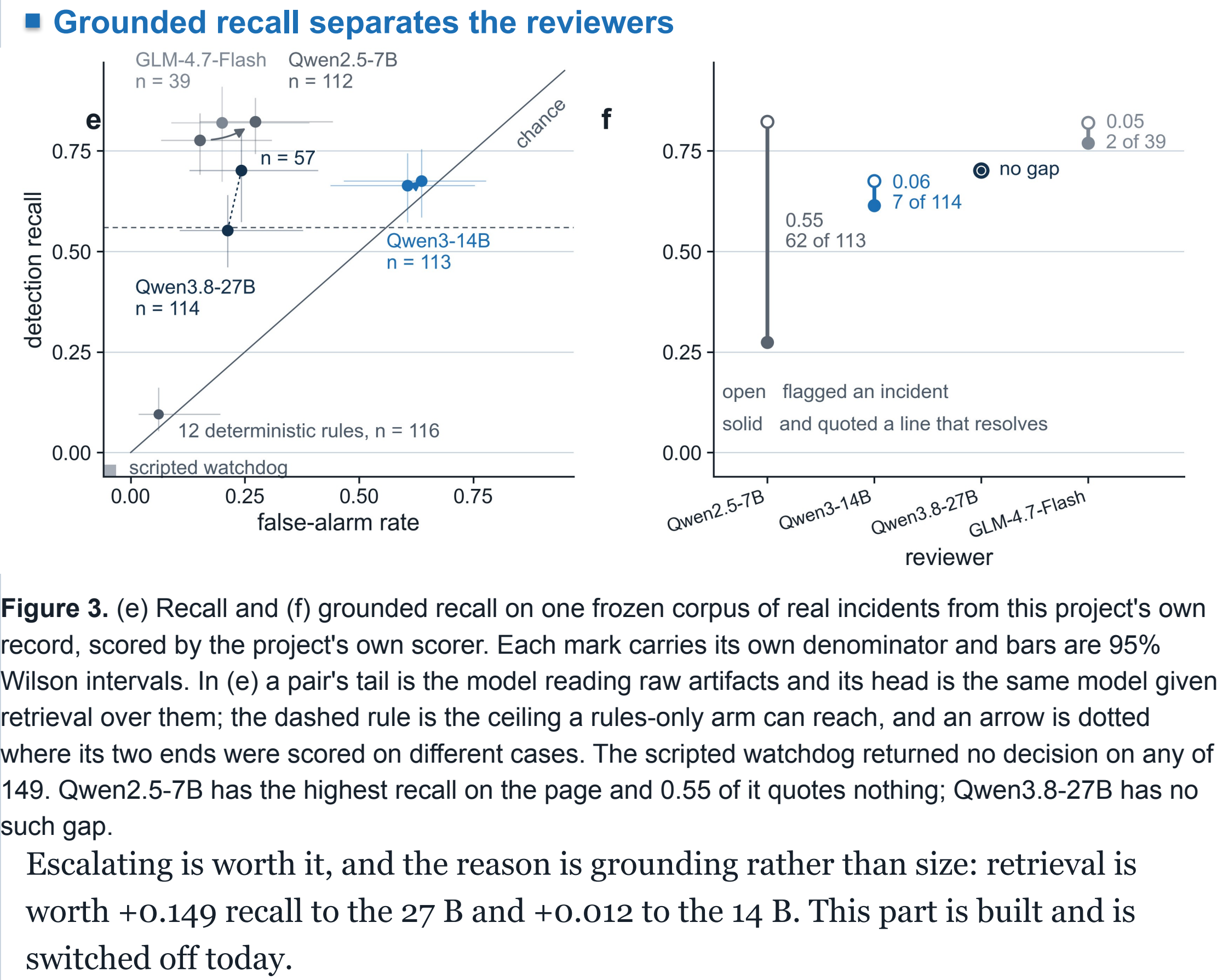


Figure 2. The loop an agent runs to answer a need, drawn as a loop. A need arrives, the agent works out what data would answer it, goes and gets it, trains on what survives the audit and scores it, then reads its own result and decides what to ask for next. When it cannot decide it climbs the ladder inside the loop, each rung reading the one below it and free to overrule it. The outward run is solid because it runs unattended today. The return run is dashed because it is the one edge the platform has not closed: the decision is recorded and nothing reads it back, and closing it is the work this poster is about.

ONE PLATFORM, ANY ROBOT, ANY DATASET

- agentAI is one platform for every robot the lab owns
- A robot, a drone or a sensor rig is a project. It owns its keys, its people and a locked dataset registry, and the agents harvest, filter, label and train against that registry. Adding a robot is a configuration change.
- Weed detection is one project here, the humanoid is still empty
- Open a project for any data type and any field, upload, and get an automatic diagnosis before choosing what to do next. Weed detection is one project here; so are the rover and the cart.
- Tracked rover and laser cart are the two robots reporting in
- Robot 241 is a tracked rover that streams camera, GPS and two IMUs. The laser cart carries a galvanometer laser, a down camera over the work zone and a forward camera.

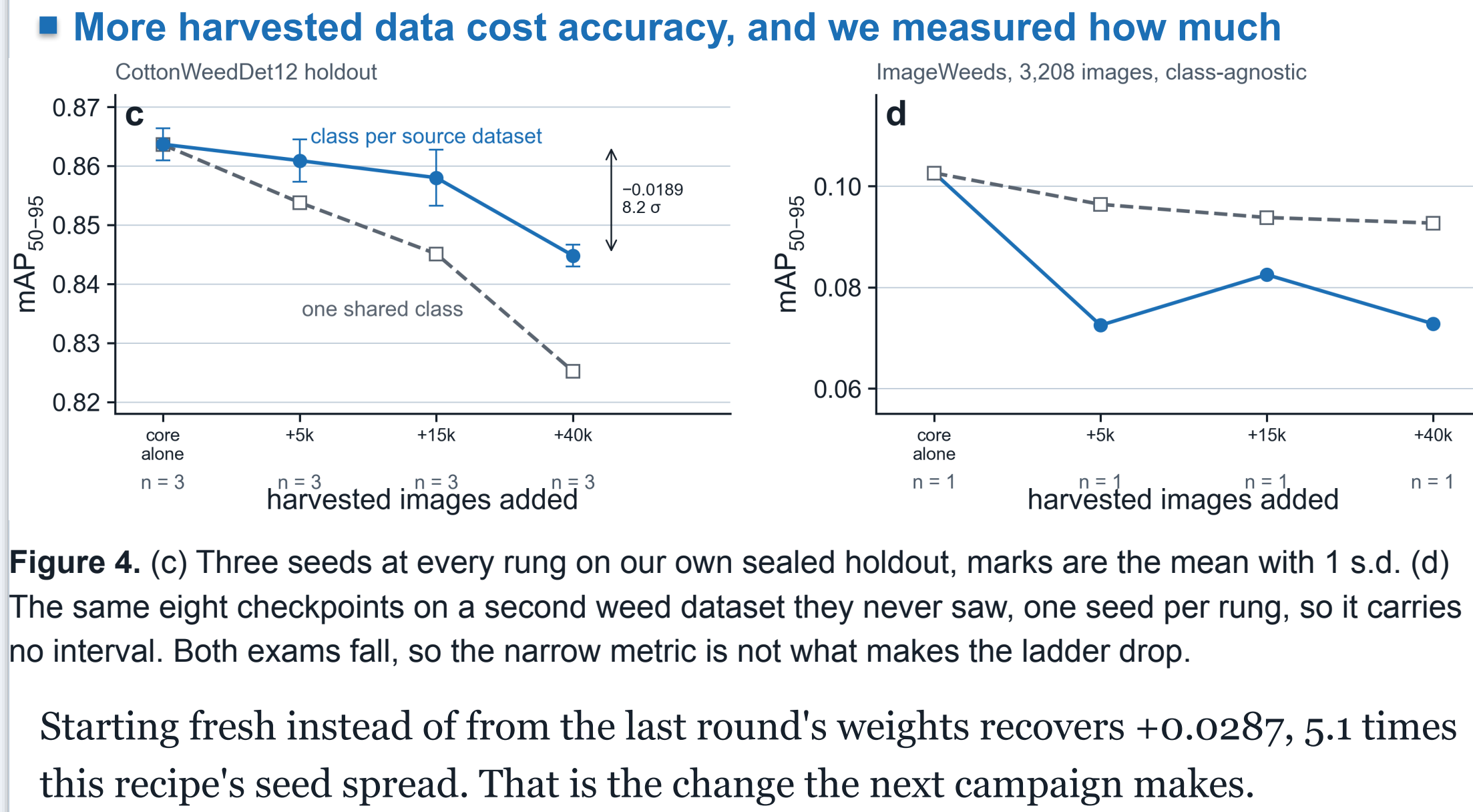
A BRAIN THAT GETS STUCK AND ASKS UPWARD



THE AGENTS RAN THE LOOP WITH NOBODY IN THE ROOM

- Fifteen rounds ran with nobody in the room
- Two cluster jobs, one collecting and one training, ran fifteen rounds against the locked registry with nobody in the room. Every round reported success. This part is built and is switched off today.
- The corpus stopped growing at round three
- The corpus stopped growing at round three: the same 24 datasets went in every time, and the apparent growth was one staging directory nobody cleared.
- Each finished drive is diagnosed without anyone asking
- Every finished drive is diagnosed without being asked -- route, noise, cross-sensor correlation, a timeline -- for 26 live robot sessions and 45 charts so far.
- One hypothesis held, the other was ruled out
- Both hypotheses were run as controls on round 15's exact data and seeds. The chain holds at 5.1 sigma; finishing the cosine schedule is worth -0.0058 at 1.3 sigma, inside the noise.

MORE DATA STOPPED HELPING, SO WE WENT AFTER THE LABELS



END STATE IS ONE PLATFORM FOR ANY DATASET AND ANY ROBOT

The end state is one general platform: any dataset, any research field, any robot. Agents on it run without being asked, iterate on their own output, escalate to a larger model when they are out of their depth, and are fenced in a sandbox while they do it. Every piece of this poster is a step in that direction. This is the direction, not a description of what runs today.

CAPABILITIES LAND IN THE PLATFORM, SO THE NEXT STUDENT INHERITS THEM

Each capability is built into the platform itself and stays there after the experiment that needed it. The routing table, the ingest contract, the dataset registry, the sandbox and the scorer are all part of the product, so the next student inherits them working. That is the difference between a lab that has done an experiment and a lab that has built somewhere to do the next one.